Sound and vibration in a fuel-oil tank – fluid loading

A pump in a tank of fuel oil sets both the oil and the tank walls vibrating, and part of the energy escapes as airborne sound. When the wall has oil on one side, it behaves quite differently from when it is surrounded only by air: the oil moves with it and makes the wall “heavier”, the natural frequencies fall, and the radiation changes. That was the subject of my individual course at the Department of Process Engineering. The page has two interactive figures on fluid-loaded plates.

> DTH, individual course at the Department of Process Engineering (1988), 6 points:
> Measurements of sound and vibration inside and outside a tank of fuel oil with a pump, and how fluid loading of the tank walls affected their vibration and radiation. The investigation supported another student’s thesis project.

The course

In industry and energy supply, tanks, vessels and pipes with pumps are everywhere – from fuel-oil storage tanks in power stations and on ships to dairies, breweries and chemical plants. They are often a significant source of noise, and the vibrations can lead to fatigue of welds and mountings. An individual course at DTH was an independent study with a supervisor at a department; this one took the form of an experimental investigation intended to provide data and understanding for another student’s thesis project. The task was to measure the sound and vibration both inside the oil and outside the tank, and to understand what the oil does to the walls’ vibration and sound radiation. This page goes through the physics behind it.

The pump as a source

A centrifugal pump produces pressure pulsations in the liquid at the blade-passing frequency $f = N_{\text{blades}} \cdot n$ and its harmonics, where n is the revolutions per second, plus a broadband contribution from turbulence and possibly cavitation. At the same time motor and pump casing send vibrations out through the foundation and pipe connections. The energy thus reaches the tank walls by three paths: as *fluid-borne sound* through the liquid, as *structure-borne sound* through metal and mountings, and as airborne sound from outside. One aim of measuring inside and outside is to find out which path dominates – for that decides which noise control will work (see machinery noise).

Vibrating plates

A tank wall is, to a first approximation, a plate vibrating with bending waves. Their speed depends on frequency – bending waves are dispersive:

$$c_B = \sqrt{\omega}\left(\frac{B}{m''}\right)^{1/4}, \qquad B = \frac{E h^3}{12(1-\nu^2)}, \qquad m'' = \rho_p h$$

For a rectangular, simply supported plate with sides a and b the natural frequencies are

$$f_{mn} = \frac{\pi}{2}\sqrt{\frac{B}{m''}}\left[\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2\right]$$

and the mode shape is $\sin(m\pi x/a)\sin(n\pi y/b)$. At each natural frequency the plate can vibrate strongly, and this is where the pump's tones are dangerous: if a blade-passing frequency falls close to a natural frequency, the vibrations become large.

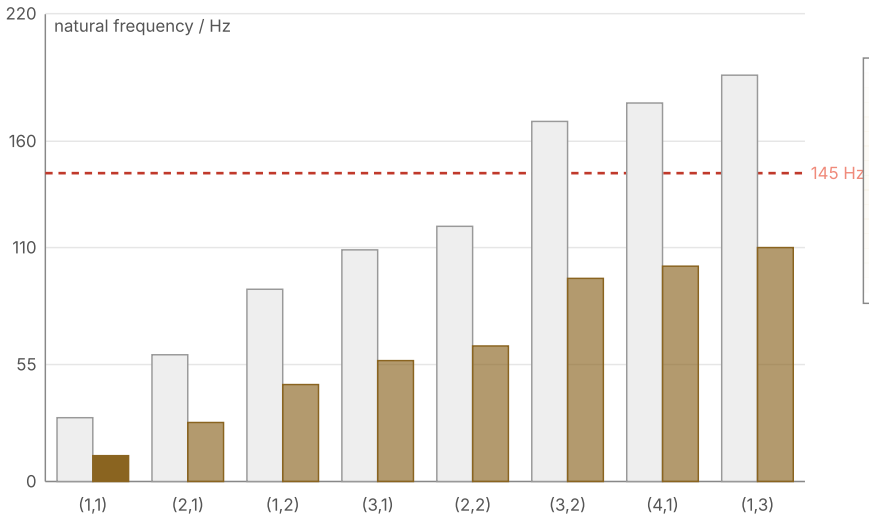
Fluid loading

When the plate vibrates, it must push the surrounding medium back and forth. In air this is negligible, but oil is about 700 times denser than air (water 800 times), and a vibrating tank wall must move some of the oil with it. Below the coincidence frequency, where the bending waves are slower than sound in the liquid, the liquid cannot radiate the energy away; instead a layer of liquid moves with the wall as an *added mass*. For a mode with wavenumber k_p in the plate the added mass per unit area (for $k_p \gg k$ in the liquid) is roughly ρ_f / k_p per loaded side, and the natural frequency falls to

$$f_{\text{wet}} \approx \frac{f_{\text{dry}}}{\sqrt{1 + \dfrac{\rho_f}{m''\,k_p}}}, \qquad k_p = \pi\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

The lowest modes, which have long wavelengths and hence small k_p, are affected most – in a thin steel tank of oil their frequency can fall to less than half. The liquid also damps the vibrations, and when the tank is closed the wall modes couple with the liquid's own acoustic modes (standing waves in the liquid) into one combined system. That is why an empty and a full tank sound so different when you knock on them.

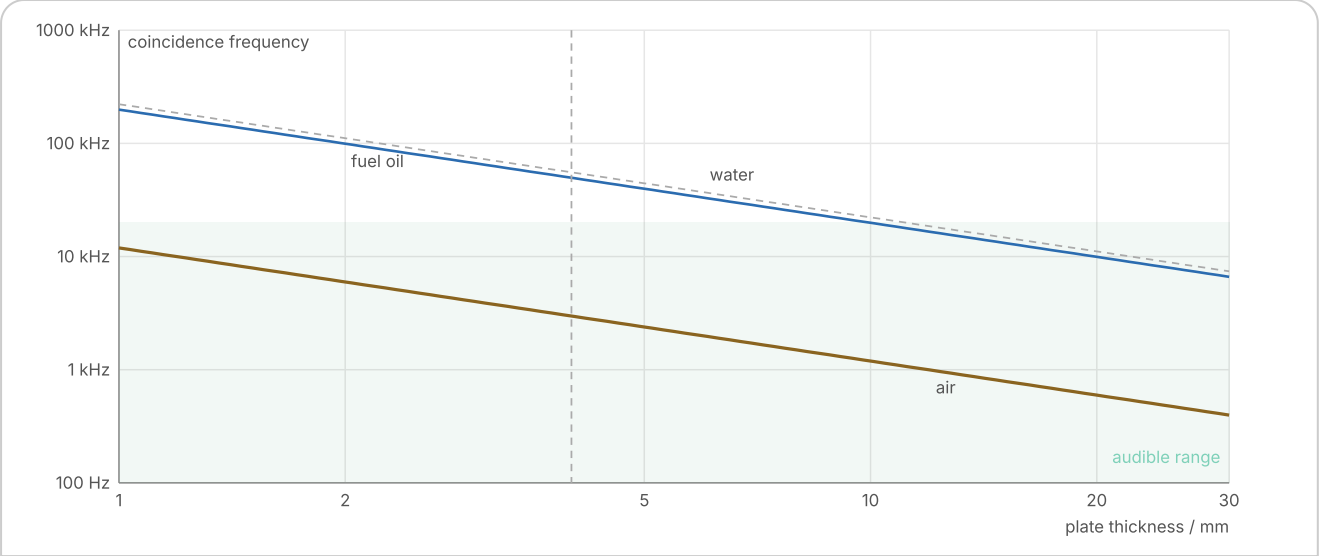
Plate in air and oil



Radiation and coincidence

How much sound a vibrating plate radiates is described by the radiation efficiency $\sigma = W/(\rho c S \langle v^2 \rangle)$: the ratio of the radiated power to that which a large piston with the same mean velocity would radiate. Above the coincidence frequency, where the bending waves are faster than sound in the medium, $\sigma \approx 1$ – the plate radiates efficiently. Below coincidence, neighbouring regions vibrating in antiphase cancel, and only edges and corners radiate; σ is much less than 1.

The coincidence frequency is $f_c = \frac{c^2}{2\pi}\sqrt{\frac{m''}{B}}$ and depends on the square of the speed of sound in the medium. Sound travels at about 1400 m/s in fuel oil – a little over 4 times faster than in air – so the coincidence frequency on the oil side is about 17 times higher. A 4 mm steel plate has $f_c \approx 3$ kHz towards air, but about 50 kHz towards oil. On the oil side the plate is thus almost always below coincidence: it radiates poorly into the oil, but the oil loads it heavily. On the air side it is the other way round – above 3 kHz the same plate radiates efficiently into the room. That is why it matters to measure both inside and outside.



Measurements inside and outside

Inside the oil the sound is measured with hydrophones – piezoelectric pressure transducers that can be immersed in the liquid. On the walls the vibrations are measured with accelerometers, and outside the tank the sound pressure or sound intensity with microphones. With a dual-channel analyser one can find transfer functions and coherence between the pump's vibrations, the pressure in the oil, the wall acceleration and the sound outside (see signal analysis). From the measured acceleration and sound power the radiation efficiency of the wall can be determined, and by comparing the empty and the full tank one sees the effect of fluid loading directly: the natural frequencies move down, and the ratio of vibration to radiated sound changes.

The measurements must be interpreted with care. The accelerometer's own mass loads a thin plate; the position of the hydrophone relative to the standing waves in the oil matters a great deal; and the airborne sound outside includes both sound from the tank and from the pump's motor.

Since 1989

- **Coupled FEM/BEM:** Structure and liquid are now computed together with finite element models of the tank and boundary element or FEM models of the liquid – the same technique is used for submarines, ship hulls and aircraft fuel tanks.
- **Laser vibrometry:** Mode shapes are measured without contact by scanning laser Doppler vibrometers, so that the mass of accelerometers does not interfere.
- **Condition monitoring of pumps:** Vibration and hydrophone measurements are used continuously to detect cavitation and wear before the pump fails.
- **Seismic and sloshing analysis:** The motion of the liquid in large tanks during earthquakes – a low-frequency version of the same coupling between liquid and wall – is now a standard part of storage-tank design.

Further reading

- Frank Fahy and Paolo Gardonio: *Sound and Structural Vibration: Radiation, Transmission and Response* (2nd ed., Academic Press 2007)
- Lothar Cremer, Manfred Heckl and Björn A. T. Petersson: *Structure-Borne Sound* (3rd ed., Springer 2005)
- M. C. Junger and D. Feit: *Sound, Structures, and Their Interaction* (2nd ed., MIT Press 1986)

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